

Srinith Nangunoori

B.Tech, Computer Science and Engineering
Sardar Vallabhbhai National Institute of Technology (SVNIT)

+91-8790788928

✉ srinith.nangunoori@gmail.com

LinkedIn | GitHub

LeetCode | CodeForces | AtCoder

EDUCATION

- **Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat** 2024 – 2028 (Expected)
Bachelor of Technology in Computer Science and Engineering **CGPA: 9.02 / 10.0**
- **New Vision Junior College** 2022
Board of Intermediate Education, Telangana **Percentage: 94.1%**

TECHNICAL SKILLS

Languages: C, C++17, Python, JavaScript, SQL, HTML/CSS
Core Systems: Multithreading & Concurrency (`std::shared_mutex`), Memory Management, TCP/IP Sockets & WebSockets, REST API Design
Frameworks/Tools: React.js, Node.js, Express.js, Electron.js, AWS EC2, ESP32/Arduino, Streamlit, Git/Linux
Databases: PostgreSQL, MongoDB (Mongoose), Custom In-Memory Data Stores
Auth & AI Integration: JWT, bcrypt, Google Gemini API
Languages Known: English (Fluent), Japanese (JLPT N4 – conversational proficiency), Telugu (Native), Hindi (Conversational)

COMPETITIVE PROGRAMMING

- **Algorithmic Problem Solving:** Solved **600+** problems across LeetCode, CodeForces, and AtCoder, with a focus on Graph Theory, Dynamic Programming, and Data Structures.

PROJECTS

- **LiteDB: High-Frequency Distributed Fraud Detection Engine** [Live Demo](#) | [Git Repo](#)
C++17, Raw TCP/WebSockets, Node.js, React, AWS EC2
 - Built a multi-threaded C++17 in-memory storage engine from first principles, bypassing relational-DB overhead to process transactions in under 50 microseconds.
 - Designed a double-buffered, asynchronous Write-Ahead Log using `std::swap`-based pointer-swapping to decouple disk I/O from the live transaction thread, and added periodic RDB snapshotting for crash recovery.
 - Used **C++17 `std::shared_mutex`** read/write locks to allow unlimited concurrent reads while serializing writes, eliminating contention under concurrent load.
 - Implemented $O(1)$ fraud heuristics: Bloom filters (FNV-1a/DJB2 hashing) for blacklist checks, Haversine geospatial math for impossible-travel-velocity detection, and bipartite-graph clustering with a Bayesian ratio threshold to flag merchant-level laundering rings while protecting high-volume merchants from false positives.
 - Bridged the engine to a Node.js orchestrator over a dual-socket TCP gateway (separating transactional and query traffic) and built a GPU-accelerated React dashboard streaming live telemetry over WebSockets.
- **Plate2Purpose: Surplus Food Redistribution Platform** [Live Demo](#) | [Git Repo](#)
MongoDB, Express.js, React.js, Node.js, JWT
 - Architected a full-stack MERN platform connecting Donors, NGOs, and Volunteers with role-based dashboards and document-based user verification.
 - Modeled a multi-stage donation lifecycle (Available → Claimed → In-Progress → Delivered) with synchronized state across linked FoodPost and Delivery documents.
 - Implemented secure, stateless authentication using JWT and bcrypt, with global Axios interceptors for token handling.
- **Synapse Health: AI-Powered Wellness Consultation Tool** [Live Demo](#) | [Git Repo](#)
React.js, Node.js, Google Gemini API, Vercel
 - Designed a two-stage interactive AI consultation flow: an initial symptom funnel feeds a preliminary analysis, which the AI uses to generate personalized clarifying questions before a refined final assessment.
 - Engineered a resilient Node.js backend with a multi-model fallback system to maintain high availability of the Gemini API integration.
- **LogiBot: Autonomous AI Warehouse Navigator** [Video Demo](#) | [Git Repo](#)
C++, Python, A Search Algorithm, ESP32*
 - Built a distributed Goal-Based Rational Agent, separating reflexive edge control (ESP32 firmware for line-following and obstacle avoidance) from centralized strategic planning (Python server).

- Implemented **A* Informed Search** with a Manhattan Distance heuristic to compute optimal, obstacle-aware paths across a 13-node RFID-localized topological graph, with real-time dynamic rerouting on path blockage.

●Integrated Business System: Manufacturing ERP Desktop App

[Git Repo](#)

Electron.js, Node.js, PostgreSQL (NeonDB)

- Designed a normalized (3NF) relational schema for a manufacturing business spanning employees, departments, inventory, suppliers, and buyers, and implemented it as a cross-platform Electron desktop application.
- Built transactional workflows (purchases, sales, and production runs) with automatic, constraint-enforced stock adjustments across linked inventory tables, plus a live dashboard and full transaction history/audit view.

LEADERSHIP & ACHIEVEMENTS

●**Leadership:** Executive Member, **ACM Student Chapter, SVNIT.**

●**CodeCraft, MindBend'26 (SVNIT's Techno-Managerial Fest):** Competed in a 4-round technical event (quiz screening, code golf, live debugging, and a final escape-room round) as part of a 3-member team; advanced through to the Round 3 Debugging Challenge.

●**HackVortex Codestorm 5 (Health & Wellness Hackathon):** Built **Synapse Health** (see Projects) as a submission to this hackathon focused on educational, health-and-wellness-oriented tools.